Problem 2

(a)

Each of the M = 302 frames (500 × 500, grayscale) is flattened into a column of length 250000, giving a 250000 × 302 matrix. The mean frame is subtracted from every column and the reduced SVD $X = U S V^T$ is computed with `full_matrices=0`, so U is 250000 × 302 rather than 250000 × 250000.

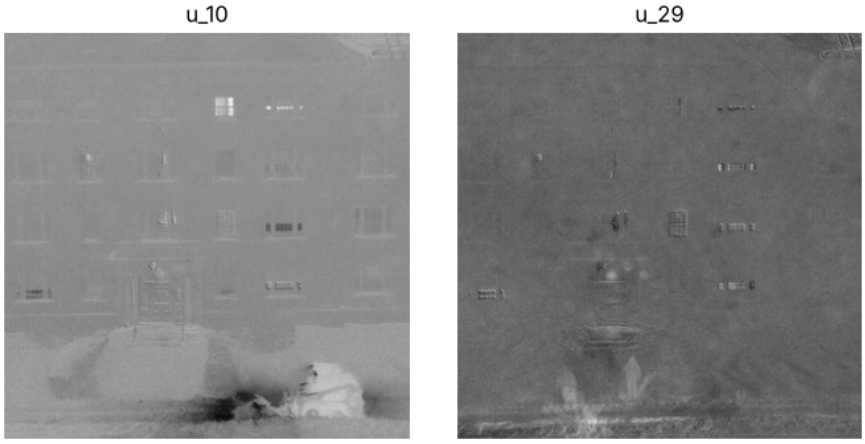
```
trainingDorms[:, i] = img.flatten()
avgDorm = np.mean(trainingDorms, axis=1)[:, np.newaxis]
img = np.reshape(avgDorm, (m, n))
X = trainingDorms - np.tile(avgDorm, (1, M))
U, S, VT = np.linalg.svd(X, full_matrices=0)
```

The first four left singular vectors, reshaped to 500 × 500:



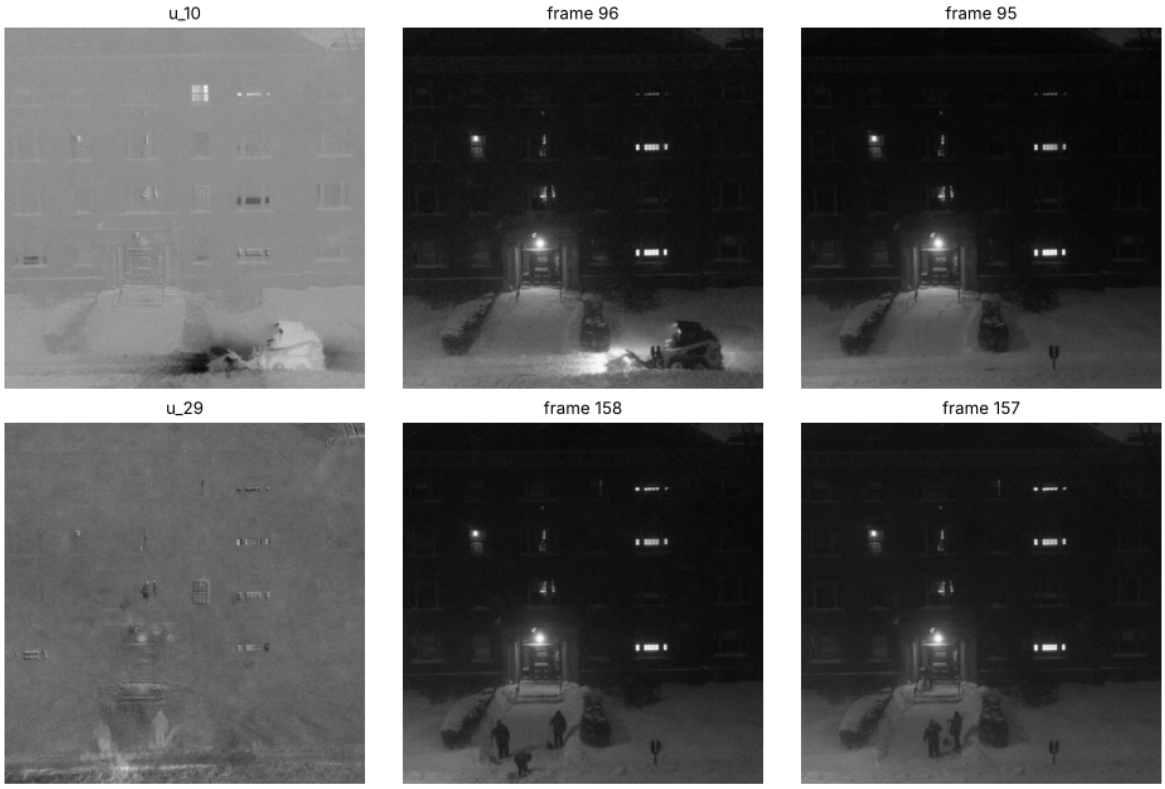
$\mathbf{u}_1$ is the outline of the whole building with the lit windows and entrance; $\mathbf{u}_2$ isolates one very bright window; $\mathbf{u}_3$ is an overall brightness pattern of the facade and the snow-covered ground; $\mathbf{u}_4$ is a single column of windows on the right.

(b)

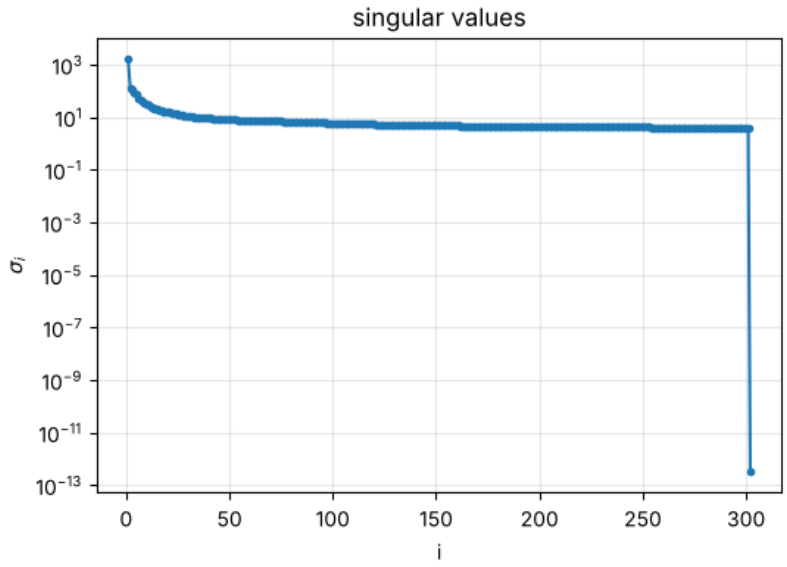


Both contain transient events rather than lighting patterns. $\mathbf{u}_{10}$ shows a bright object in the lower right corner: its time coefficient $v_{10}(t)$ is dominated by a single frame (frame 96, coefficient −0.87), and that frame shows a snow plow driving past the entrance. $\mathbf{u}_{29}$ shows

several human figures in front of the entrance; its largest coefficient is at frame 158 (–0.38), followed by frames 18 and 159; frames 158–159 show people shoveling snow in front of the entrance, which is where the figures come from. The SVD puts events that appear in only a few frames into later singular vectors, since they explain only a small share of the total variance.

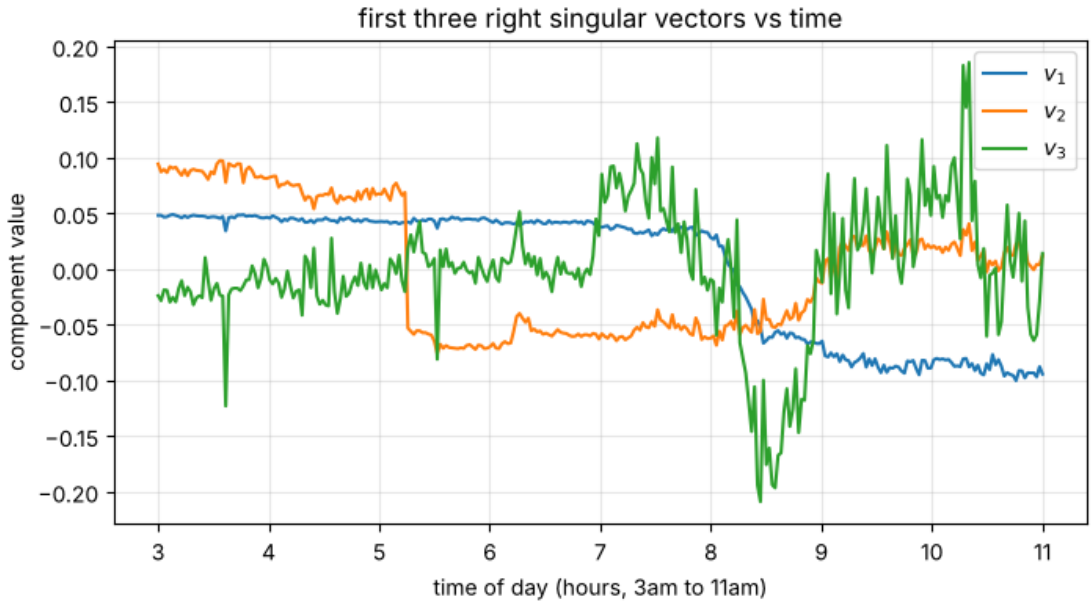


(c)



σ_1 = 1638.6 is about 12 times larger than σ_2 = 132.7; after that the singular values decay slowly into a long tail of single-digit values, and the last one is zero because the mean-subtracted columns sum to zero (rank X = 301). The cumulative energy $\Sigma_{i\leq k}\sigma_i^2 / \Sigma\sigma_i^2$ is 0.976 for k = 1, 0.992 for k = 5, 0.995 for k = 10 and 0.996 for k = 20. Given the sharp drop after the first singular value and the flattening after about 10, keeping roughly 5 to 10 singular vectors is a reasonable low-rank approximation for compression; going beyond ~20 buys very little.

(d)



The 302 frames span 3am to 11am, so each $v_k(t)$ is the weight of the pattern $\mathbf{u}_k$ at time t. $\mathbf{v}_1$ is nearly constant and positive until about 8am and then switches to negative between 8 and

9am: this is sunrise, the single largest change in the movie, when the whole scene goes from dark with lit windows to daylight. $\mathbf{v}_2$ has a sharp step at about 5:15am and a second change near 8:30am; combined with $\mathbf{u}_2$, which isolates one bright window, the step corresponds to that window's light being switched off. $\mathbf{v}_3$ is small and noisy at night and fluctuates strongly between 7 and 9am, the period with the most activity in the scene (people shoveling, changing snowfall and illumination).

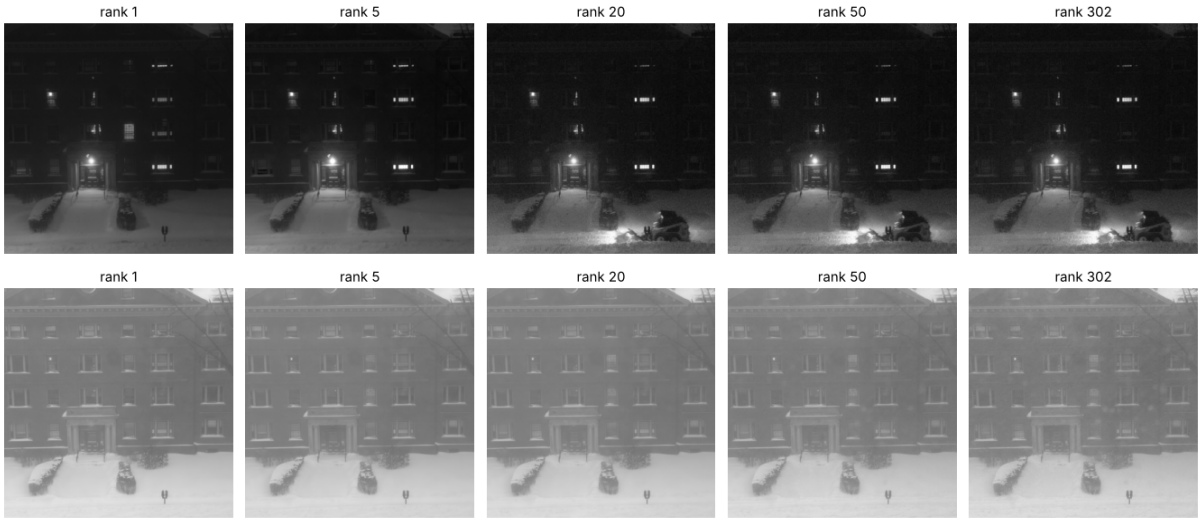
(e)

The rank-k approximation of the movie is $X_k = U_k S_k V_k^T$ (first k columns of U, first k singular values, first k rows of V^T), and the frames are the columns of X_k plus the mean frame. `play(k)` displays the rank-k movie for any k from 1 to M.

```
def lowrank_movie(k):
    Xk = U[:, :k] @ (S[:k, np.newaxis] * VT[:k, :])
    return Xk + avgDorm                      # pixels x M, column t = frame t

def play(k):
    frames = lowrank_movie(k)
    ax = None
    for i in range(M):
        img = np.reshape(frames[:, i], (m, n))
        if ax is None:
            ax = plt.imshow(img, cmap=plt.cm.gray, vmin=0, vmax=1)
        else:
            ax.set_data(img)
        plt.pause(.001); plt.draw()
```

Frame 96 (night, snow plow) and frame 250 (daylight) at ranks 1, 5, 20, 50 and 302:



Rank 1 reproduces only the mean building plus the day/night change; the snow plow is missing. It appears at rank 20 because $\mathbf{u}_{10}$ is then included. Ranks 50 and 302 are visually indistinguishable. The relative Frobenius error $||X - X_k|| / ||X||$ is 0.155 (k = 1), 0.091 (k = 5), 0.072 (k = 10) and 0.060 (k = 20), while a rank-10 movie needs only 3.3% of the original storage.